DOCUMENT 07

Database Specification

PostgreSQL / Supabase schema, RLS, indexes, retention, and ER diagram

Study Topic Extractor

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Turn long study material into a focused study roadmap.

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1. Scope and Principles

The database is **Phase 2**. The MVP has no database: results live in the client session and the server keeps only an in-memory cache. This schema is designed now so the MVP result JSON maps cleanly onto it later, and so the schema supports every Phase 2 feature.

- PostgreSQL 15 on Supabase; migrations with Alembic; SQLAlchemy 2.0 models mirror this document exactly.
- UUID primary keys (`gen_random_uuid()`), `timestamptz` timestamps in UTC, `snake_case` names.
- Deleting a document or user cascades to all derived data (privacy requirement).
- The backend enforces ownership in code; Row Level Security is enabled as defense in depth.
- Analysis output is immutable: re-analysis creates a new `analysis_runs` row.

2. Entity-Relationship Diagram

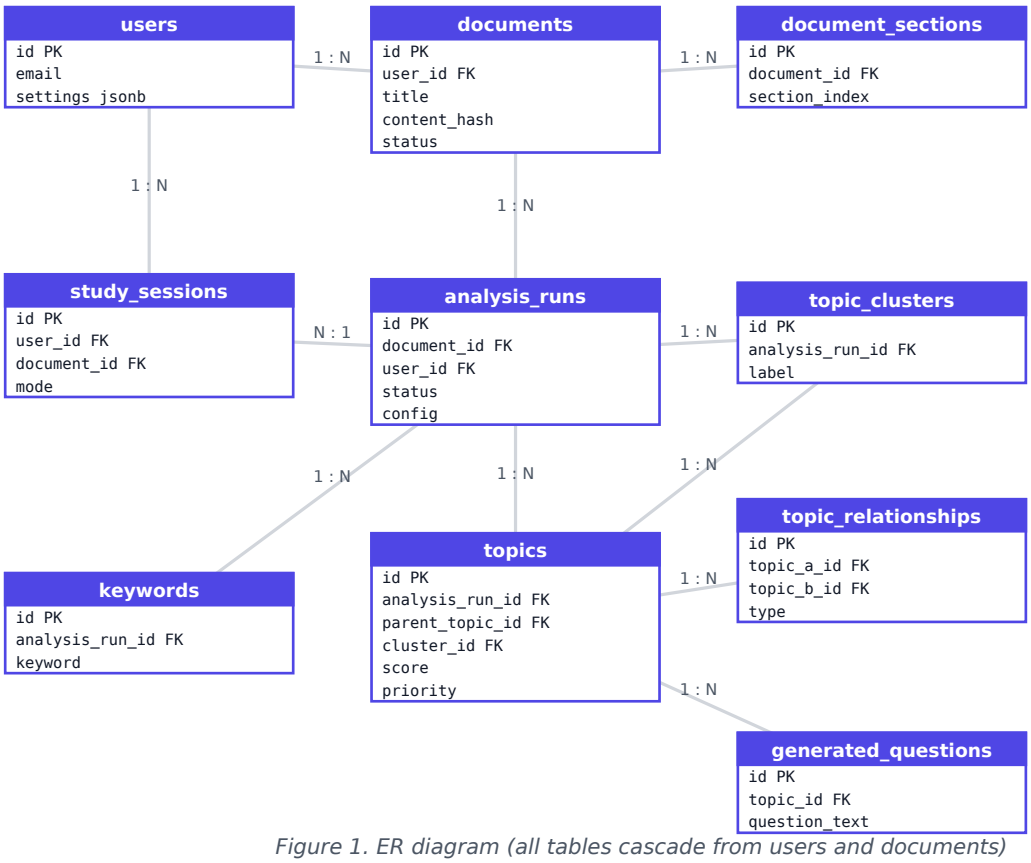


Figure 1. ER diagram (all tables cascade from users and documents)

Relationship	Cardinality	Delete behavior
users to documents	1 : N	CASCADE
documents to document_sections	1 : N	CASCADE
documents to analysis_runs	1 : N	CASCADE
analysis_runs to topics / keywords / topic_clusters / topic_relationships	1 : N	CASCADE
topic_clusters to topics	1 : N	SET NULL on topics.cluster_id
topics to topics (parent)	1 : N (self)	SET NULL on parent_topic_id
topics to generated_questions	1 : N	CASCADE
users and documents to study_sessions	1 : N	CASCADE

3. Table Definitions

3.1 Table: users

Application profile mirroring Supabase `auth.users`. Created on first sign-in.

Column	Type	Key / constraint	Null	Default	Description
`id`	uuid	PK, FK auth.users(id)	No	-	Same id as the Supabase auth user
`email`	text	UNIQUE	No	-	Lowercased e-mail
`display_name`	text	-	Yes	NULL	Optional name
`settings`	jsonb	-	No	'{}'	Theme, default mode, max_topics, weights, thresholds
`created_at`	timestamptz	-	No	now()	Creation time
`updated_at`	timestamptz	-	No	now()	Updated by trigger

3.2 Table: documents

One uploaded or pasted study document owned by a user.

Column	Type	Key / constraint	Null	Default	Description
`id`	uuid	PK	No	gen_random_uuid()	Document id
`user_id`	uuid	FK users(id) ON DELETE CASCADE	No	-	Owner
`title`	text	-	No	'Untitled document'	Max 120 chars (CHECK)
`source_type`	text	CHECK in ('text','pdf')	No	-	Origin
`file_path`	text	-	Yes	NULL	Storage key `{user_id}/{document_id}.pdf` for PDFs
`file_size_bytes`	integer	CHECK >= 0	Yes	NULL	PDF size
`content_hash`	char(64)	-	No	-	SHA-256 of normalized extracted text; duplicate detection
`extracted_text`	text	-	No	-	Cleaned text used for analysis
`char_count`	integer	CHECK 200..150000	No	-	Length
`word_count`	integer	-	No	-	Word count
`page_count`	integer	-	Yes	NULL	PDF pages
`extraction_method`	text	CHECK in ('paste','native','ocr')	No	'paste'	How text was obtained

Column	Type	Key / constraint	Null	Default	Description
extraction_quality	real	CHECK 0..1	Yes	NULL	PDF extraction quality
language	text	-	No	'en'	Detected language code
status	text	CHECK in ('uploaded','extracted','analyzing','analyzed','failed')	No	'extracted'	Lifecycle
created_at	timestamptz	-	No	now()	Creation
updated_at	timestamptz	-	No	now()	Update
expires_at	timestamptz	-	Yes	NULL	Optional retention deadline

3.3 Table: document_sections

Detected sections or windows of a document.

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Section id
document_id	uuid	FK documents(id) ON DELETE CASCADE	No	-	Parent
section_index	integer	UNIQUE(document_id, section_index)	No	-	Order, from 0
heading	text	-	Yes	NULL	Heading text
heading_level	smallint	CHECK 1..6	Yes	NULL	Level
start_char	integer	-	No	-	Start offset in extracted_text
end_char	integer	CHECK > start_char	No	-	End offset
page_start	integer	-	Yes	NULL	First page
page_end	integer	-	Yes	NULL	Last page
token_count	integer	-	No	-	Tokens

3.4 Table: analysis_runs

One execution of the pipeline for a document.

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Run id (equals analysis_id in API)
document_id	uuid	FK documents(id) ON DELETE CASCADE	No	-	Analyzed document
user_id	uuid	FK users(id) ON DELETE CASCADE	No	-	Owner (denormalized for RLS)

Column	Type	Key / constraint	Null	Default	Description
status	text	CHECK in ('queued','running','succeeded','failed','cancelled')	No	'queued'	Job state
config	jsonb	-	No	-	Weights, thresholds, flags, max_topics
pipeline_version	text	-	No	-	Version string
model_versions	jsonb	-	No	'{}'	spaCy, MiniLM, background IDF hash
stats	jsonb	-	No	'{}'	topics_found, high, medium, low, warnings
error_code	text	-	Yes	NULL	Error catalog code
processing_ms	integer	-	Yes	NULL	Duration
started_at	timestamptz	-	Yes	NULL	Start
finished_at	timestamptz	-	Yes	NULL	End
created_at	timestamptz	-	No	now()	Creation

3.5 Table: topic_clusters

Semantic clusters of topics within a run.

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Cluster id
analysis_run_id	uuid	FK analysis_runs(id) ON DELETE CASCADE	No	-	Run
cluster_index	integer	UNIQUE(analysis_run_id, cluster_index)	No	-	Order
label	text	-	No	-	Highest-scoring member name
algorithm	text	CHECK in ('agglomerative','kmeans','dbscan')	No	'agglomerative'	Method
size	integer	CHECK >= 1	No	-	Members
cohesion	real	-	Yes	NULL	Mean intra-cluster cosine

3.6 Table: topics

Ranked topics of a run.

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Topic id

Column	Type	Key / constraint	Null	Default	Description
analysis_run_id	uuid	FK analysis_runs(id) ON DELETE CASCADE	No	-	Run
document_id	uuid	FK documents(id) ON DELETE CASCADE	No	-	Document
cluster_id	uuid	FK topic_clusters(id) ON DELETE SET NULL	Yes	NULL	Cluster
parent_topic_id	uuid	FK topics(id) ON DELETE SET NULL	Yes	NULL	Hierarchy parent
name	text	-	No	-	Display name
normalized_name	text	UNIQUE(analysis_run_id, normalized_name)	No	-	Lemma key
rank	integer	CHECK >= 1	No	-	1 = best
score	numeric(4,1)	CHECK 0..100	No	-	Importance score
priority	text	CHECK in ('HIGH','MEDIUM','LOW')	No	-	Priority label
ngram_size	smallint	CHECK 1..3	No	-	Words in topic
occurrences	integer	CHECK >= 1	No	-	Count
first_position_pct	real	CHECK 0..1	No	-	First occurrence position
components	jsonb	-	No	-	tfidf, frequency, position, semantic
summary	text	CHECK length <= 400	No	-	Extractive or hybrid summary
definition	text	-	Yes	NULL	Definition sentence
keywords	text[]	-	No	'{}'	Per-topic keywords
depth	smallint	CHECK 0..4	No	0	Hierarchy depth
created_at	timestamptz	-	No	now()	Creation

3.7 Table: keywords

Document-level keywords of a run.

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Keyword id
analysis_run_id	uuid	FK analysis_runs(id) ON DELETE CASCADE	No	-	Run
document_id	uuid	FK documents(id) ON DELETE CASCADE	No	-	Document
keyword	text	-	No	-	Display form

Column	Type	Key / constraint	Null	Default	Description
normalized_keyword	text	UNIQUE(analysis_run_id, normalized_keyword)	No	-	Lemma key
score	real	CHECK 0..1	No	-	Normalized score
occurrences	integer	-	No	-	Count
ngram_size	smallint	CHECK 1..3	No	-	Words

3.8 Table: topic_relationships

Edges between topics (hierarchy, semantic, cooccurrence).

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Edge id
analysis_run_id	uuid	FK analysis_runs(id) ON DELETE CASCADE	No	-	Run
topic_a_id	uuid	FK topics(id) ON DELETE CASCADE	No	-	Source (parent for hierarchy)
topic_b_id	uuid	FK topics(id) ON DELETE CASCADE	No	-	Target (child for hierarchy)
relationship_type	text	CHECK in ('hierarchy','semantic','cooccurrence')	No	-	Edge type
similarity_score	real	CHECK 0..1	No	-	Evidence / cosine / normalized count
Table constraints		UNIQUE(analysis_run_id, topic_a_id, topic_b_id, relationship_type); CHECK topic_a_id <> topic_b_id			

3.9 Table: generated_questions

Possible questions per topic.

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Question id
topic_id	uuid	FK topics(id) ON DELETE CASCADE	No	-	Topic
document_id	uuid	FK documents(id) ON DELETE CASCADE	No	-	Document
question_text	text	-	No	-	Question
question_type	text	CHECK in ('definition','explanation','list','comparison','why_how','application')	No	-	Template type

Column	Type	Key / constraint	Null	Default	Description
difficulty	text	CHECK in ('easy','medium','hard')	No	'medium'	Difficulty
generation_method	text	CHECK in ('template','llm')	No	'template'	Method
source_sentence	text	-	Yes	NULL	Grounding sentence
created_at	timestamptz	-	No	now()	Creation

3.10 Table: study_sessions

Exam Focus or study activity records.

Column	Type	Key / constraint	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Session id
user_id	uuid	FK users(id) ON DELETE CASCADE	No	-	Owner
document_id	uuid	FK documents(id) ON DELETE CASCADE	No	-	Document
analysis_run_id	uuid	FK analysis_runs(id) ON DELETE CASCADE	No	-	Run studied
mode	text	CHECK in ('quick','normal','deep')	No	-	Study mode
reviewed_topic_ids	uuid[]	-	No	'{}'	Topics marked reviewed
started_at	timestamptz	-	No	now()	Start
ended_at	timestamptz	-	Yes	NULL	End
duration_seconds	integer	CHECK >= 0	Yes	NULL	Length

4. Indexes

Index	Table (columns)	Purpose
idx_documents_user_created	documents (user_id, created_at DESC)	History listing
uq_documents_user_hash	documents (user_id, content_hash) UNIQUE	Duplicate detection
idx_documents_expires	documents (expires_at) WHERE expires_at IS NOT NULL	Retention job
idx_sections_doc	document_sections (document_id, section_index)	Section lookup
idx_runs_doc_created	analysis_runs (document_id, created_at DESC)	Latest run
idx_runs_user_status	analysis_runs (user_id, status)	Job polling
idx_topics_run_rank	topics (analysis_run_id, rank)	Ordered topic list
idx_topics_run_priority	topics (analysis_run_id, priority)	Priority filter
idx_topics_parent	topics (parent_topic_id)	Tree traversal
idx_keywords_run_score	keywords (analysis_run_id, score DESC)	Top keywords
idx_rel_run_type	topic_relationships (analysis_run_id, relationship_type)	Graph load
idx_questions_topic	generated_questions (topic_id)	Question lookup
idx_sessions_user_started	study_sessions (user_id, started_at DESC)	Dashboard stats

5. Constraints, Triggers, and Cascading

- All CHECK constraints listed in the table definitions are created in the initial migration.
- `set_updated_at()` trigger on `users` and `documents`.
- Deleting a `documents` row deletes: sections, runs (and therefore topics, keywords, clusters, relationships, questions) and sessions. The API also deletes the Storage object; if storage deletion fails the row is still deleted and the object key is queued in a `storage_cleanup` log for retry.
- Deleting a user (`DELETE /api/me`) deletes the profile, which cascades to everything owned; then the Supabase auth user and the user's Storage prefix `{user_id}/` are removed.
- The application layer enforces `max 200 documents per user` (configurable) to control storage use.

6. Row Level Security

RLS is enabled on every table. The FastAPI backend connects with a server-side role and enforces ownership in its repository layer (every query includes `user_id`). RLS policies protect against any direct access using the public anon key or a user JWT.

```
ALTER TABLE documents ENABLE ROW LEVEL SECURITY;
CREATE POLICY documents_owner ON documents
  FOR ALL TO authenticated
  USING (user_id = auth.uid()) WITH CHECK (user_id = auth.uid());

-- Child tables inherit ownership through the document:
ALTER TABLE topics ENABLE ROW LEVEL SECURITY;
CREATE POLICY topics_owner ON topics FOR SELECT TO authenticated
  USING (EXISTS (SELECT 1 FROM documents d
                 WHERE d.id = topics.document_id AND d.user_id = auth.uid()));
-- Writes to derived tables are performed only by the backend role; no INSERT/UPDATE
-- policy is granted to `authenticated` for topics, keywords, topic_clusters,
-- topic_relationships, generated_questions.
-- Storage: bucket `documents` is private; policy allows access only to
-- objects whose first path segment equals auth.uid()::text.
```

Table	authenticated SELECT	authenticated write	anon
users	own row	update own settings/display_name	none
documents	own rows	insert/update/delete own	none
document_sections, analysis_runs, topics, keywords, topic_clusters, topic_relationships, generated_questions	via owning document	none (backend only)	none
study_sessions	own rows	insert/update own	none

7. Data Retention and Privacy

Data	Retention
Documents and derived results	Kept until the user deletes them; optional `expires_at` (Settings: keep 30, 90, 180 days or forever). Default 180 days for new documents. A daily job deletes expired rows.
Stored PDFs	Same as parent document; deleted with it.
Failed analysis runs	Deleted after 30 days.
Study sessions	Kept with the document; deleted on cascade.
Logs	No document content; retained 30 days.
Anonymous MVP usage	No content stored server-side; in-memory cache entries expire within 1 hour.
Account deletion	Immediate hard delete of all user data, completed within 24 hours including backups purge policy of the provider.

8. Migration Order and Seed Data

1. `0001_create_users_documents_sections`
2. `0002_create_analysis_runs_topics_keywords`
3. `0003_create_clusters_relationships_questions_sessions`
4. `0004_indexes_constraints_triggers`
5. `0005_rls_policies_storage`

No seed data is required. Test fixtures are created by `tests/factories.py`.

8.1 Mapping From API Result to Tables

API field	Table.column
`analysis_id`	`analysis_runs.id`
`document.*`	`documents.*`
`config`	`analysis_runs.config`
`stats`	`analysis_runs.stats`
`topics[]`	`topics` (components to JSONB, keywords to text[])
`keywords[]`	`keywords`
`hierarchy`	`topics.parent_topic_id`, depth, plus hierarchy rows in `topic_relationships`
`graph.edges[]`	`topic_relationships`
`warnings[]`	`analysis_runs.stats.warnings`

9. Acceptance Criteria

1. Alembic upgrade from empty to head succeeds; downgrade to base succeeds.
2. A user cannot read another user's rows using the anon key or a different user's JWT (verified by tests).
3. Deleting a document removes all rows in dependent tables and the storage object.
4. Inserting a topic with score 101, priority `URGENT`, or depth 5 fails.
5. The Phase 2 API can persist and reload any Analysis Result without loss of fields.